



Mst.Moushumi Khatun

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● ABOUT ME

Published AI Engineer and Research Assistant with a proven track record in Machine Learning, Deep Learning, and Advanced NLP, specializing in high-impact applications across healthcare, autonomous agent systems, and predictive analytics. Highly skilled in designing complex sequential frameworks and deep learning pipelines, with multiple peer-reviewed publications in ACM, Springer, and manuscripts Accepted in Q1 venues (PLOS ONE). Possesses hands-on expertise in Graph-based Multi-Agent Systems (MAS), reward-shaped Reinforcement Learning (RL) environments for smart grids, and Large Language Model (LLM) alignment loops. Driven to pursue a PhD under Associate Professor Jimmy Cao to pioneer next-generation, safe, and cooperative decentralized multi-agent reinforcement learning architectures.

● EDUCATION AND TRAINING

2018/10/13 – 2022/12/22 Dhaka, Bangladesh
B.SC. ENG IN COMPUTER SCIENCE & ENGINEERING Bangladesh University of Business and Technology(BUBT)

Website <https://www.bubt.edu.bd/>

● WORK EXPERIENCE

JR.AI ENGINEER – SAVE THE CHILDREN INTERNATIONAL – 2026/04/01 – Current – DHAKA , BANGLADESH

- **Human-Aligned Core Decision Systems:** Formulated and deployed a **Reinforcement Learning from Human Feedback (RLHF)** pipeline to align Large Language Models (LLMs) with ethical child welfare standards when processing Datix case reports.
- **Reward Model Engineering:** Architected and trained a specialized **Reward Model** based on explicit feedback and evaluations from domain-expert case officers and child-protection lawyers.
- **Mitigating Toxicity & Hallucinations:** Utilized the trained reward model to fine-tune generative policies via Proximal Policy Optimization (PPO), successfully reducing model hallucinations and toxic or biased outputs in sensitive child-abuse risk classifications.

RESEARCH ASSISTANT – ADVANCED MACHINE INTELIGENCE RESEARCH LAB – 2022/09/01 – Current – DHAKA, BANGLADESH

- **Algorithm Design & Model Training:** Formulated and optimized deep learning architectures (including Convolutional Neural Networks and Sequential Models) for high-precision computer vision and multi-modal predictive tasks.
- **Full-Lifecycle Data Engineering:** Architected automated data curation, multi-modal feature extraction, and complex annotation pipelines for diverse, real-world, and low-resource datasets.
- **Collaborative Research Execution:** Collaborated with cross-functional research teams to develop dual-stream deep learning systems for real-time impact scoring, alert prioritization, and automated decision-making engines.

SENIOR INSTRUCTOR-REMOTE – BRITSYNC – 2025/08/01 – Current – WALES , UNITED KINGDOM

- Designed and developed AI and Data Science courses for students and professionals
- Mentored junior team members and guided learners to achieve practical competency in AI/ML.
- Created curriculum outlines, coding exercises, and project-based assessments aligned with international AI standards..

JUNIOR DATA SCIENTIST-REMOTE – BRITSYNC – 2024/07/01 – 2025/07/31 – WALES , UNITED KINGDOM

- Developed and optimized machine learning models for AI projects. Conducted EDA to extract actionable insights, improving decision-making efficiency.
- Performed data preprocessing, cleaning, and feature engineering for diverse datasets.
- Deployed predictive Solutions Using AWS EC2 and Containerized Applications.

DATA ANALYST – SKILL SERVICE LTD – 2023/02/01 – 2024/06/30 – DHAKA

- Collected, cleaned, and organized **5,000+ image datasets** for computer vision model training.
- Performed image preprocessing (resizing, normalization, augmentation), improving model accuracy by **20%**.
- Annotated images using **Labelling, CVAT, Roboflow**, ensuring high-quality bounding boxes and segmentation masks.
- Collaborated with the AI team to build training datasets for **object detection and classification models (YOLO, CNN)**.

SKILLS

1. Programming Languages

Python PHP JavaScript SQL HTML CSS

2. Artificial Intelligence & Machine Learning

PyTorch TensorFlow Keras scikit-learn XGBoost | Convolutional Neural Networks (CNN) Long Short-Term Memory (LSTM)

3. Advanced RL & Agent Architectures

Reinforcement Learning (RL) Multi-Agent Systems (MAS) | Reward Shaping & Continuous Action Control
Reinforcement Learning from AI Feedback (RLAIF)

4. NLP & Generative AI

Large Language Models (LLMs) LangChain Hugging Face Transformers R | Retrieval-Augmented Generation (RAG)
spaCy NLTK Translation Models

5. Data Science & Business Intelligence Tools

Pandas NumPy Power BI Matplotlib Seaborn Microsoft Excel Data Visualization & Curation

6. Cloud, DevOps & Deployment

AWS (EC2) Docker Kubernetes Git / GitHub Flask Streamlit REST API Development

PROJECTS

2025/06/06 – 2025/06/07

Graph-Based Multi-Agent System for Autonomous Scientific Ideation

Engineered a decentralized network of specialized LLM agents structured as a topological graph to automate complex scientific literature synthesis and collaborative hypothesis generation. Implemented a robust agent-to-agent message passing protocol over directed graphs to enable iterative, multi-agent peer-review. Utilized dynamic knowledge graphs to capture evolving semantic relations between multi-disciplinary concepts, significantly mitigating contextual drift and hallucinations compared to single-agent baselines.

Link https://github.com/moushumipriya/Graph_based_multi_agent_systems_for_scientific_ideation

HydroPowerRL-Reinforcement-Learning-for-Hydro-Reservoir-Management

HydroPowerRL is a reinforcement learning project designed to optimize water release decisions in a hydroelectric reservoir. The goal is to maximize electricity generation revenue while maintaining reservoir safety by avoiding floods and droughts.

This project uses a Deep Q-Network (DQN) agent trained on synthetic hourly inflow and electricity price data to learn effective reservoir management strategies.

Link <https://github.com/moushumipriya/Norway-HydroPowerRL-Reinforcement-Learning-for-Hydro-Reservoir-Management>

Guardrail-Driven Victim Case Analysis & AI Recommendation System

Developed an end-to-end safeguarding system utilizing Large Language Models (LLMs) to extract sensitive risk metrics and behavioral patterns from raw victim case narratives. Built an automated Reinforcement Learning from AI Feedback (RLAIF) loop acting as an algorithmic validation guardrail. The RL engine successfully penalizes unverified outputs and rewards policy alignment, ensuring generated legal and care recommendations remain fully compliant with socio-legal frameworks.

Link <https://github.com/moushumipriya/Guardrail-Driven-Victim-Case-Analysis-AI-Recommendation-System>

PUBLICATIONS

2025

[Self-Supervised Representation Learning for Human Activity Recognition Using Inertial Sensor Data](#)

Authors: Mst. Moushumi Khatun; Mohammad Shafat Ahsan; Md Al Amin; Arifa Akter Eva; Anika Sayeema; Sabiha Firdaus | **Journal Name:** 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN) | **Publisher:** IEEE

2025

[DeepUNet24: A Deep Learning Model for Lane Segmentation in Autonomous Vehicles Navigation](#)

Authors: Hashibul Ahsan Shoaib, Mst. Moushumi Khatun, Md. Golam Rabbani Abir, Adit Ishraq, Sabiha Firdaus, Kamruddin Nur, M. Firoz Mridha | **Journal Name:** ICCA '24 | **Volume, Issue and Pages:** Pages 406 - 413 | **Publisher:** ACM

2025

[Multi-Sense Attention Network \(MSANet\): Enhanced Human Activity Recognition Using Deep Learning Architectures with Self-Attention Mechanisms](#)

Authors: Hashibul Ahsan Shoaib, Arifa Eva, Mst. Moushumi Khatun, Adit Ishraq, Sabiha Firdaus, Dr. M. Firoz Mridha | **Journal Name:** ICCA '24 | **Volume, Issue and Pages:** Pages 362 - 369 | **Publisher:** ACM

2024

[Yoga Posture Image Classification Using Big Transfer\(BiT\)](#)

Authors: K.M Nafiur Rahman Fuad; Uland Rozario; Hashibul Ahsan Shoaib; Mst. Moushumi Khatun; Md Raisul Islam; M. F. Mridha | **Journal Name:** 2024 IEEE International Conference on Computing, Applications and Systems (COMPAS) | **Publisher:** IEEE

2025

[Deep Learning Algorithms for Detecting Banana Leaf Spot Diseases](#)

Authors: Hadiur Rahman Nabil, Md. Golam Rabbani Abir, Mst. Moushumi Khatun, Md. Eshmam Rayed & Md. Abdul Hamid | **Journal Name:** Machine Vision in Plant Leaf Disease Detection for Sustainable Agriculture | **Volume, Issue and Pages:** Chapter pp 155–167 | **Publisher:** Springer

2025

[MoraleTrack: A Dual-Phase Transformer-Based Framework for Sentiment-Aware Team Morale Forecasting in Agile Project Environments](#)

Authors: Mohammad Shafat Ahsan; Md Al Amin; Ishtiaq Al Mamoon; Mst. Moushumi Khatun; Sheikh Razia Sultana; Kamruddin Nur | **Journal Name:** 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things | **Publisher:** IEEE

2025

[Multi-View Federated Learning for Cybersecurity Threat Detection Using Network and User Behavior Data](#)

Authors: Moushumi Khatun; Anika Sayeema; Mumtahina Ahmed; Arifa Akter Eva; Sheikh Razia Sultana; Sabiha Firdaus | **Journal Name:** 28th International Conference on Computer and Information Technology | **Publisher:** IEEE

Authors: Mohammed Saad Javeed; Mst. Moushumi Khatun; Jobayar Alom; Rahomotul Islam; Hashibul Ahsan Shoaib | Journal Name: PLOS ONE

CERTIFICATIONS

IT Training BD, 2024/06/01

Hands-on SQA and Cyber Security

Mode of learning: Online

Business Automation Ltd, 2025/09/04

Software Development

- Developed and maintained web modules using Laravel (PHP), HTML, CSS, and JavaScript.
- Participated in SQA processes including test case creation and bug reporting.
- Worked in Agile teams and managed code via Git.

Mode of learning: Online

Link <https://ba-systems.com/>

CONFERENCES AND SEMINARS

2026/04/11 – 2026/04/12 Dhaka

National Career Conference - Youth Skill Development Institute
